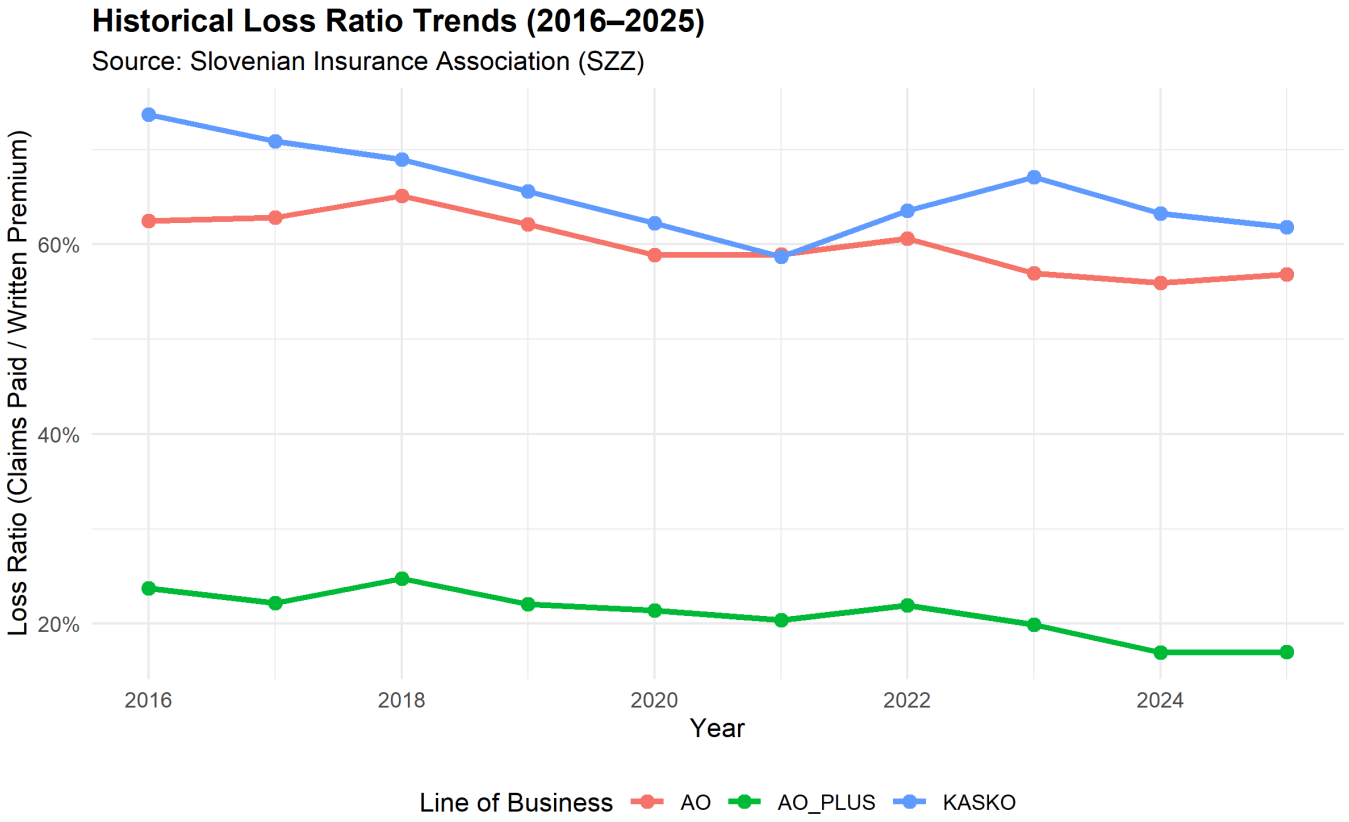
Solvency II Motor insurance Actuarial & Reserve Model

An actuarial modeling framework for analyzing the Slovenian motor insurance market (MTPL/AO, AO+, KASKO) using historical data from the Slovenian Insurance Association (SZZ) spanning 2016–2025.

The project processes market data in an SQLite database, estimates claim severity inflation via log-linear regression, and runs Monte Carlo simulations (Poisson + Lognormal) to compute Best Estimate Liabilities (**BEL**) and Solvency Capital Requirements (**SCR**) under Solvency II standards.

LIVE DEMO

Historical Loss Ratio Validation (R Output)



Core Features

- **Data Storage (SQLite):** Reads historical premiums, claims paid, and policy counts from SZZ records into a local `szz_market.db` database.
- **Claims Inflation:** Uses log-linear regression to estimate severity growth trends and underlying volatility (σ) across business lines.
- **Monte Carlo Loss Simulation:** Simulates aggregate claims for 2026 projections:
  - **Claim Frequency:** Poisson distribution (λ).
  - **Claim Severity:** Lognormal distribution (μ, σ) with lognormal convexity adjustment ($\exp(\mu + 0.5\sigma^2)$).
- **Solvency Capital ($SCR_{99.5}$):** Estimates Value at Risk at a 99.5% confidence level over a 1-year horizon.

- **R Data Validation:** Independent R script (`dplyr`, `ggplot2`) to query the database and audit historical loss ratio trends.
- **Streamlit Dashboard:** Interactive web interface for adjusting simulation parameters and visualizing loss distributions.

Repository Structure

```
.
├── data/                  # Historical CSV data (SZZ records 2016–2025)
├── szz_market.db          # Local SQLite database (generated)
├── database_setup.py      # Database setup and data ingestion script
├── model_engine.py        # Actuarial calculations and Monte Carlo simulation
├── app.py                 # Streamlit dashboard
├── scripts/
│   └── analysis_module.R  # R validation script
├── requirements.txt       # Python dependencies
└── README.md              # Documentation
```

How It Works

- **Database Setup (`database_setup.py`):** Loads raw historical CSV data into SQLite (`szz_market.db`).
- **Actuarial Calculations (`model_engine.py`):**
  - Extracts exposure, claims, and loss ratios from the database.
  - Fits log-linear regression to estimate claim severity inflation.
  - Projects 2026 expected liabilities (BEL) with lognormal convexity adjustments.
  - Runs Monte Carlo loss simulations (2,000+ iterations) across portfolio segments.
- **Interactive Dashboard (`app.py`):** Displays aggregated portfolio projections and tail risk metrics.
- **Independent R Audit (`scripts/analysis_module.R`):** Queries SQLite directly to verify historical loss ratios and generate plots with `ggplot2`.

Key Projections (2026)

- **Best Estimate Liabilities (BEL):** Total projected claims of €637.0M across all three motor lines for 2026.
- **Solvency Capital Requirement ($SCR_{99.5}$):** Capital buffer estimated at €44.8M at a 99.5% Value at Risk level to absorb 1-year tail risk.
- **Portfolio Exposure:** Comprehensive Motor (KASKO) and MTPL (AO) represent the largest market shares, with historical 10-year average loss ratios of 65.6% and 60.1%, respectively.

Key Takeaways & Design Decisions

- **Data Management:** Decouples raw CSV storage from modeling logic using an SQLite database layer.
- **Actuarial Modeling:** Uses compound distributions (Poisson claim counts + Lognormal claim amounts) matching standard non-life reserving practices.

- **Regulatory Standard:** Aligns capital adequacy metrics ($VaR_{99.5}$) with Solvency II standard formula guidelines.
- **Cross-Validation:** Uses R alongside Python to cross-check loss ratio metrics directly against the database.

Limitations & Future Enhancements

- **Reserving Triangles (IBNR):** The current model relies on historical annual aggregates. Integrating claim development run-off triangles will enable traditional actuarial reserving methods (e.g., Chain Ladder, Bornhuetter-Ferguson) to capture Incurred But Not Reported (IBNR) liabilities.
- **Simulation Granularity (Individual Claims vs. Aggregates):** Portfolio losses are generated by multiplying simulated annual Poisson counts with simulated average Lognormal severities. Implementing individual claim simulation (or Panjer recursion) will ensure mathematically exact aggregate compound loss distributions.
- **Inter-Line Dependency (Copulas):** The framework assumes independence across business lines (MTPL, AO+, KASKO). Implementing copula functions (e.g., Gaussian, Student-t, or Clayton) will model correlated tail risk during systemic events like severe weather shocks.
- **Macroeconomic Inflation Scenarios:** Severity inflation is estimated via a constant log-linear trend. Expanding sensitivity testing to evaluate dynamic macroeconomic shocks (e.g., CPI spikes, supply-chain auto part inflation) will improve capital adequacy stress testing.

How to Run the Project

1. **Clone the repository:**

```
git clone https://github.com/klepecvabicrok/Solvency-II-Motor-insurance-Actuarial-Reserve-Model-2026.git
cd Solvency-II-Motor-insurance-Actuarial-Reserve-Model-2026
```

2. **Install dependencies:**

```
pip install -r requirements.txt
```

3. **Initialize the database:**

```
python database_setup.py
```

4. **Launch the dashboard:**

```
streamlit run app.py
```

5. **Run the R audit script:**

```
Rscript scripts/analysis_module.R
```